

Yiwei Chen

 Github |  Homepage |  LinkedIn |  Google Scholar

EDUCATION

Michigan State University

Ph.D. in Computer Science, Advisor: Prof. [Sijia Liu](#)

Aug. 2024 – Present
East Lansing, United States

Xi'an Jiaotong University

M.S. in Computer Science and Technology

Sept. 2021 – Jun. 2024
Xi'an, China

Xi'an Jiaotong University

B.Eng. in Automation, [Young Gifted Class](#) (national early-entrance honors program)

Sept. 2017 – Jun. 2021
Xi'an, China

RESEARCH INTERESTS

- **Agents & Post-training:** LLMs, MLLMs, Post-training, Reasoning, Agentic Systems, MAS, GUI Agents
- **Trustworthy AI:** AI Safety, Alignment, Trustworthy Algorithms, Interpretability, Machine Unlearning

INTERNSHIPS

Amazon, Applied Scientist Intern

Research Project: Memory-Augmented World Models for Mobile GUI Agents

May 2026 – Aug. 2026

Sunnyvale, United States

- Identified the loop-closure problem in GUI world models, where one-step predictors regenerate a revisited page instead of restoring its earlier state, and formulated a trajectory-level prediction task with loop-closure evaluation.
- Built an automated annotation pipeline adding page identity, HTML targets, and revisit labels to three benchmarks.
- Designed a latent memory that stores hidden states of each observed transition, retrieves them when a page is revisited, and injects them into a frozen VLM via zero-initialized cross-attention trained with counterfactual losses.
- Outperformed prior GUI world models on revisit consistency; manuscript in preparation for CVPR 2027.

Cisco, Research Intern

Research Project: LLMs for CVE Exploit Generation in Cybersecurity [\[M.1\]](#)

May 2025 – Oct. 2025

San Francisco, United States

- Built a CVE-conditioned exploit generation dataset and an 8-criterion LLM-as-judge evaluation framework.
- Benchmarked 17 open-weight and proprietary LLMs under one setup to characterize their zero-shot capability.
- Fine-tuned Qwen3-8B on the curated data, improving exploit quality by 42.5% and rivaling proprietary models.

PUBLICATIONS

C=Conference, M=Manuscript, *=Equal Contribution

- [C.1] [Y. Chen](#), C. Wang, Y. Lang, Q. Qu, A. O.Hero, S. Liu. [Train Locally, Learn Together: Distributed Multi-Agent Training via ADMM](#). *In Submission*.
- [C.2] [Y. Chen](#), Y. Yao, Y. Zhang, B. Shen, G. Liu, S. Liu. [Safety Mirage: How Spurious Correlations Undermine VLM Safety Fine-Tuning and Can Be Mitigated by Machine Unlearning](#). **ICLR'26**.
- [C.3] [Y. Chen](#), S. Pal, Y. Zhang, Q. Qu, S. Liu. [Unlearning Isn't Invisible: Detecting Unlearning Traces in LLMs from Model Outputs](#). **ICLR'26**.
- [C.4] [Y. Chen](#), B. Shang, S. Liu. [Who Built This Model? Tracing LLM Lineage via Spectral Fingerprints in Weight Space](#). **COLM'26**.
- [C.5] [Y. Chen](#)^{*}, B. Shang^{*}, Y. Zhang, B. Shen, S. Liu. [Forgetting to Forget: Attention Sink as A Gateway for Backdooring LLM Unlearning](#). **COLM'26**.
- [M.1] [Y. Chen](#), L. Li, K. Cheung, V. Parla, G. Sundaram. [Data-Centric Benchmarking of Exploit Generation in LLMs: Understanding the Impact of Fine-Tuning](#). Manuscript, 2026.
- [C.6] [Y. Chen](#)^{*}, Z. Zhang^{*}, W. Zhang, C. Yan, Q. Zheng, Q. Wang, W. Chen. [Tile Classification Based Viewport Prediction with Multi-modal Fusion Transformer](#). **ACM MM'23**.
- [C.7] H. Reisizadeh, J. Ruan, [Y. Chen](#), S. Pal, S. Liu, M. Hong. [Leak@k: Unlearning Does Not Make LLMs Forget Under Probabilistic Decoding](#). **ICML'26**.
- [C.8] Y. Zhang, C. Wang, [Y. Chen](#), C. Fan, J. Jia, S. Liu. [The Fragile Truth of Saliency: Improving LLM Input Attribution via Attention Bias Optimization](#). **NeurIPS'25 (Spotlight)**.

RESEARCH EXPERIENCE

OPTML Group, Michigan State University <i>Research Assistant, Advisor: Prof. Sijia Liu</i>	Aug. 2024 – Present <i>East Lansing, United States</i>
◦ <i>Trustworthy Foundation Models</i>: Investigated the reliability of post-training interventions: characterized spurious correlations in VLM safety fine-tuning and proposed an unlearning-based remedy [C.2]; established that unlearning leaves detectable output-level traces [C.3] and is vulnerable to attention-sink backdoors [C.5]; introduced spectral weight-space fingerprints for model lineage attribution [C.4]. ◦ <i>Multi-Agent Systems</i>: Developed a distributed training pipeline for multi-agent LLM systems whose agents reside on separate nodes without checkpoint sharing; cast it as consensus optimization and derived an ADMM algorithm that synchronizes only low-rank adapter states while provably matching centralized training [C.1]. ◦ <i>GUI Agents</i>: Investigated long-horizon planning and memory for GUI agents, with a focus on maintaining consistent decision making across extended multi-step interactions with mobile interfaces.	

HONORS AND AWARDS

Scholarships & Awards	
International Conference on Learning Representations, Travel Award	Apr. 2026
Michigan State University, Graduate Travel Award	Oct. 2025
Xi'an Jiaotong University, Outstanding Master Graduate (Top 5%)	Jun. 2024
Xi'an Jiaotong University, Outstanding Bachelor Graduate (Top 5%)	Jun. 2021
Xi'an Jiaotong University, First Prize Scholarship (Top 10%)	2016 – 2019, 2022 – 2023
Competitions	
Mathematical Contest in Modeling, Meritorious Winner (Top 8%)	Mar. 2019
China Undergraduate Mathematical Contest in Modeling, First Prize (Shaanxi Division)	Oct. 2018

SERVICES

• Conference Reviewer: ICLR, NeurIPS, ACM MM, ICASSP, IJCNN • Journal Reviewer: Journal of Machine Learning Research (JMLR), IEEE Transactions on Signal Processing (TSP)	
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SKILLS

• Programming: Python, C/C++, Bash, LaTeX • LLM Training & Inference: PyTorch, Hugging Face (Transformers, PEFT, TRL), DeepSpeed, verl, vLLM • Models & Agents: Llama, Qwen, Qwen-VL, LLaVA, LangGraph, MCP • Tools: Git, Docker, Slurm, Claude Code, Cursor, Codex • Languages: Mandarin (Native), English (Professional)	
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